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AIML intern

DAY 2

Today, I learned to implement pose detection through my pc webcam using OpenCV and MediaPipe in VS Code. The python code which I used was:

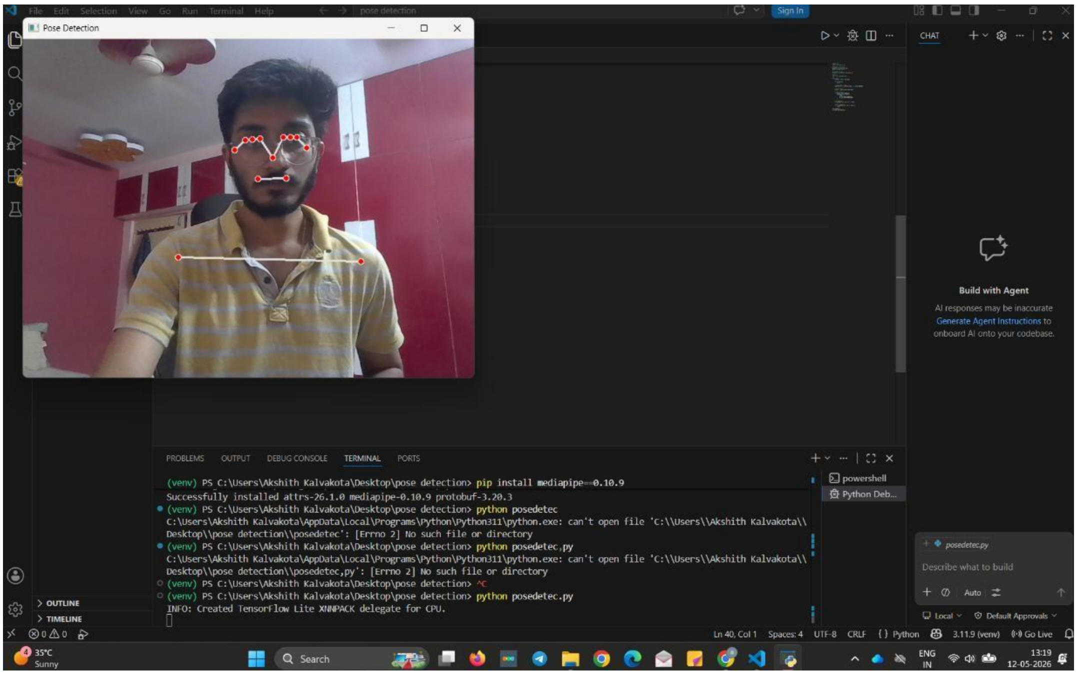
```python
import cv2
import mediapipe as mp

mp_pose = mp.solutions.pose
pose = mp_pose.Pose()

mp_draw = mp.solutions.drawing_utils

cap = cv2.VideoCapture(0)
while True:
    success, frame = cap.read()
    if not success:
        break

    rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    results = pose.process(rgb_frame)

    if results.pose_landmarks:
        mp_draw.draw_landmarks(
            frame,
            results.pose_landmarks,
            mp_pose.POSE_CONNECTIONS
        )
    cv2.imshow("Pose Detection", frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```

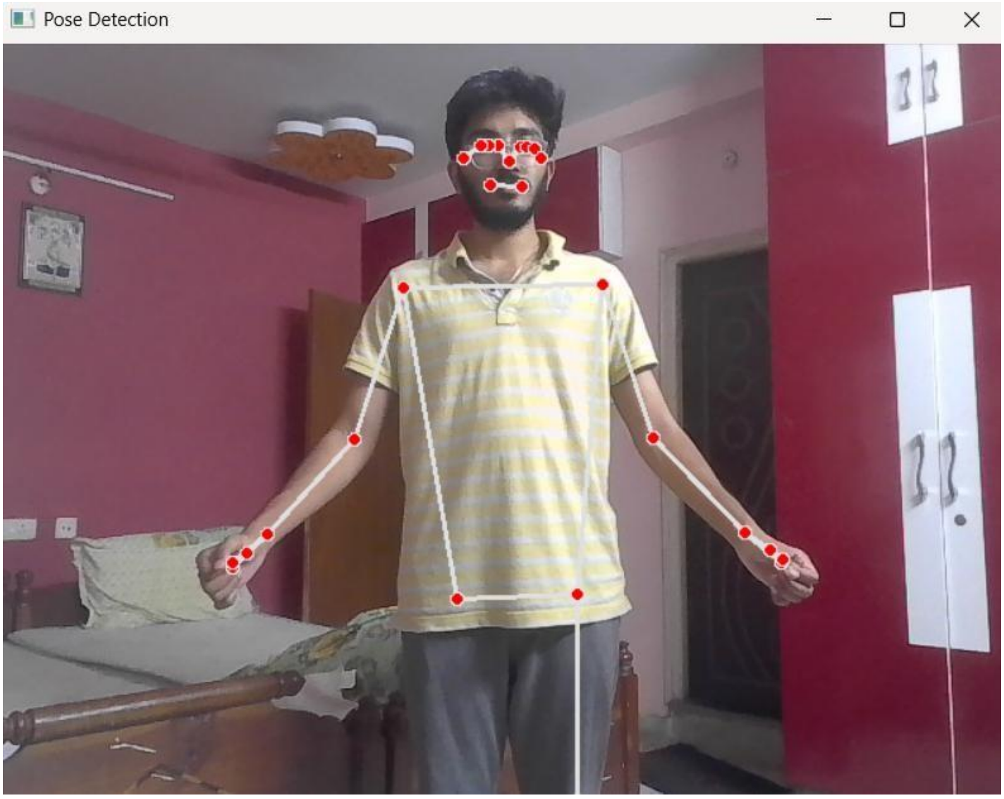
Executing this program opened the webcam and instantly showed body landmarks(points) around specific joints.

In total it shows 33 landmarks which includes:

- Nose
- Eyes (left/right)
- Ears (left/right)
- Mouth region
- Shoulders (left/right)
- Elbows (left/right)
- Wrists (left/right)
- Chest / mid-shoulders alignment
- Hips (left/right)
- Knees (left/right)
- Ankles (left/right)
- Heel (left/right)
- Foot index (left/right)

The output I got:





This is just a demo but it is very beneficial to calculate body measurements for accurate Virtual Try-On experience.

Mediapipe has built in models to automatically track the human pose landmarks which makes it the best framework to use for the Raritone application. The motion capture feature is also used in many VR videogames and athletes also use these to analyze and measure their sports abilities to improve further.

THE ISSUES I FACED:

I have two python versions on my VS Code namely 3.11.9(best for AI and Computer Vision libraries) and 3.14.2(Latest version of python). This caused a clash between the versions and hence showing that MediaPipe isn't downloaded at all. I couldn't understand the issue at first was trying to debug and tried to uninstall and reinstall MediaPipe multiple times.

WHAT I LEARNED:

After looking into it I analyzed and understood the use of a virtual environment(.venv). This helped me create a separate environment just for my project that includes the libraries it requires. I learned that I can even use the same virtual environment for other projects if it has the same tools and frameworks requirement or else for a new .venv of a new project I have to download them again.

<u>TO OPEN A VIRTUAL ENVIRONMENT:</u> *python -m venv venv*

<u>TO ACTIVATE THE VIRTUAL INTERNSHIP</u>: *venv\Scripts\Activate*

CONCLUSION:

In conclusion, on day 2 of the Raritone internship, I got hands-on experience on pose detection implementation demo using OpenCV and MediaPipe. Eagerly looking forward to learn more about these concepts and how they are implemented in real life applications.